



Optional Accessory: Wall
Cover-Up Plate F61-2663,
6 3/4" W x 4 1/2" H



INDEX

Thermostat Installation	2-4
Wiring	2
Installer Menu	3-4
Test Equipment	4
Using the Thermostat	5-7
Thermostat Overview	5
Thermostat Operation	6
Thermostat Schedule/Programming	6-7
Troubleshooting	7-8
Homeowner Help Line	8

Thermostat Applications	Maximum Stages Heat/Cool
Conventional Gas, Oil, Electric (mV and 24V), Heat Only, Cool Only or Heat/Cool Systems	1/1
Heat Pump (air source or geothermal) with no Aux. Heat	1/1

ATTENTION: MERCURY NOTICE

This product does not contain mercury. However, this product may replace a product that contains mercury. **Mercury and products containing mercury must not be discarded in household trash. Do not touch any spilled mercury.** Wearing non-absorbent gloves, clean up any spilled mercury and place in a sealed container. For proper disposal of a product containing mercury or a sealed container of spilled mercury, place it in a suitable shipping container. Refer to www.thermostat-recycle.org for location to send the product containing mercury.

SPECIFICATIONS

Electrical Rating:

Battery Power	mV to 30 VAC, NEC Class II, 50/60 Hz
Input-Hardwire	20 to 30 VAC, NEC Class II, 50/60 Hz
Terminal Load	1.5A per terminal, 2.5A maximum all terminals combined
Setpoint Range	45° to 99° F (7° to 37° C)
Rated Differentials:	Fast Med Slow
Heat (@ 6°F/ Hr).....	0.5°F 0.75°F 1.9°F
Cool (@ 6°F/ Hr).....	0.9°F 1.2°F 1.7°F
Operating Ambient	32°F to +105°F (0° to +41°C)
Display Temperature Range	32°F to +99°F (0 to 37°C)
Operating Humidity	90% non-condensing max
Shipping Temperature Range	-20°F to + 150°F (-29° to +65°C)
Thermostat Dimensions.....	3-3/4" H x 6" W x 1-1/8" D

PART NO. 37-7841003

Replaces 37-7841002

2222

THERMOSTAT INSTALLATION

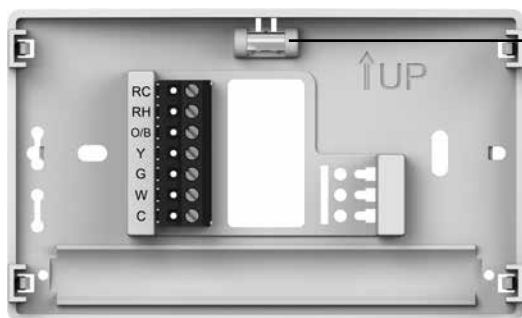
WIRING

Refer to equipment manufacturer's instructions for specific system wiring information. After wiring, see INSTALLER MENU for proper thermostat configuration. Wiring table shown are for typical systems and describe the thermostat terminal functions.

RC*	Power for Cooling
RH*	Power for Heating
O/B	Changeover Terminal-Energized in Heat (B) or Cool (O) for Heat Pump or Damper Systems
Y**	Cooling Relay
G	Fan Relay
W**	Heating Relay
C	Common wire for 24V (optional with batteries)

*When both RC and RH wires are present, cut RC/RH jumper (see next page).

**For heat pump systems, add a jumper wire to connect terminals Y and W



Leveling Thermostat
Leveling is for appearance only and will not affect thermostat operation.

Precautions

- Do not exceed the specification ratings.
- All wiring must conform to local and national electrical codes and ordinances.
- This control is a precision instrument, and should be handled carefully. Rough handling or distorting components could cause the control to malfunction.

⚠ WARNING

Do not use on circuits exceeding specified voltage.
Higher voltage will damage control and could cause shock or fire hazard.

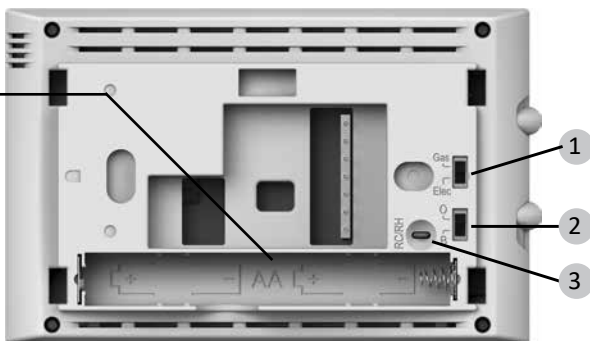
Do not short out terminals on gas valve or primary control to test. Short or incorrect wiring will burn out thermostat and could cause personal injury and/or property damage.

⚠ CAUTION

To prevent electrical shock and/or equipment damage, disconnect electrical power to system, at main fuse or circuit breaker box, until installation is complete.

Battery Location

Premium AA alkaline batteries are required when C-wire is not available. When C-wire is available, the batteries provide a back-up source of power (this will maintain the clock in the event of a power outage).



1.) Gas/Elec Switch

If the system is a heat pump or electric furnace, the GAS/ELEC Switch must be set to **Elec**. If your system is a gas or oil furnace, the switch must be set to **Gas**.

2.) O/B Terminal Switch

The O/B switch on this thermostat is factory set to the **O** position. This will accommodate the majority of heat pump applications, which require the changeover relay to be energized in **Cool**. If the heat pump being installed requires a **B** terminal, to energize the changeover relay in **Heat**, the O/B switch must be moved to the **B** position.

3.) RC/RH Jumper Wire

This thermostat electrically connects the RC and RH terminals so a jumper wire is not required. If the application provides a separate wire for RC and RH, clip the RC/RH jumper. This will isolate both terminals so they can be independently used.

INSTALLER MENU

To access the INSTALLER'S MENU set the system switch to the OFF position and then press and hold the temperature ▲ and ▼ buttons for 3 seconds. The display will show item **30** in the table below. Use the temperature ▲ and ▼ buttons by pressing them simultaneously to navigate through menu items. Press ▲ or ▼ to change a menu setting.

Installer's Menu # (Hold Menu 3 Seconds)	Description	Default Setting (flashing icons)	Settings (Press ▲ or ▼)
30 CR	Heat Cycle Rate (how often the heat will turn on)	MEd	SLO – slow MEd – medium FAS – fast
35 CR	Cool Cycle Rate (how often the cooling will turn on)	MEd	SLO – slow MEd – medium FAS – fast
50 CL	Compressor Lockout (protects the compressor from short cycling)	OFF	On – 5 minute display OFF – no delay
65	Maximum Heat Limit (maximum set point for heat mode)	99	47 to 99
66	Minimum Cool Limit (minimum set point for cool mode)	45	45 to 97
74	Schedule Type (set as either 7-Day, 5-1-1 Day or Non-Programmable)	5	7 – 7 Day 5 – 5-1-1 Day 0 – Non Programmable

(Installer Menu continued on next page)

INSTALLER MENU (Continued)

Installer's Menu # (Hold Menu 8 Seconds)	Description	Default Setting (flashing icons)	Settings
76 	Early Start (starts heating or cooling early so your programmed temperature is reached by the programmed time)	OFF	On – start early OFF – start at program period time
79	Fahrenheit or Celsius	°F	°F – Fahrenheit °C – Celsius
81	Temperature Display Adjustment (adjust the displayed "Room Temperature")	0	-5 to +5
83 	Continuous Display Light (keep the backlight always on – "C" wire required)	OFF	On – always on OFF – momentarily
86	Change Air Filter (set up a monthly reminder)	OFF	1 to 12 – reminder time (months) OFF – no filter reminder

TEST EQUIPMENT

Turn on power to the system.

Fan Operation

If your system does not have a G terminal connection, skip to **Heating System**.

- 1.) Move fan switch to On position. The blower should begin to operate.
- 2.) Move fan switch to Auto position. The blower should stop immediately.

Heating System

- 1.) Move **System** Switch to **Heat** position.
- 2.) Press ▲ to adjust thermostat setting to 1° above room temperature. The system should begin to operate and the thermostat will indicate **Heat On**.
- 3.) Press ▼ to adjust thermostat setting 1° below room temperature. The heating system should stop operating and the thermostat should indicate **Heat**.

Cooling System

- 1.) Move **System** Switch to **Cool** position.
- 2.) Press ▼ to adjust thermostat setting 1° below room temperature. The blower should come on immediately on high speed, followed by cold air circulation. The thermostat will indicate **Cool On**. There can be up to a 5 minute delay. (see INSTALLER MENU, item 50)
- 3.) Press ▲ to adjust thermostat setting to 1° above room temperature. The cooling system should stop operating and the thermostat will indicate **Cool**.

*Note: If **Starting Soon** is shown on the display, the compressor lockout feature is operating. There will be up to a 5 minute delay before the compressor turns on. (see INSTALLER MENU, item 50)*



CAUTION

To prevent compressor and/or property damage, if the outdoor temperature is below 50°F, DO NOT operate the cooling system.

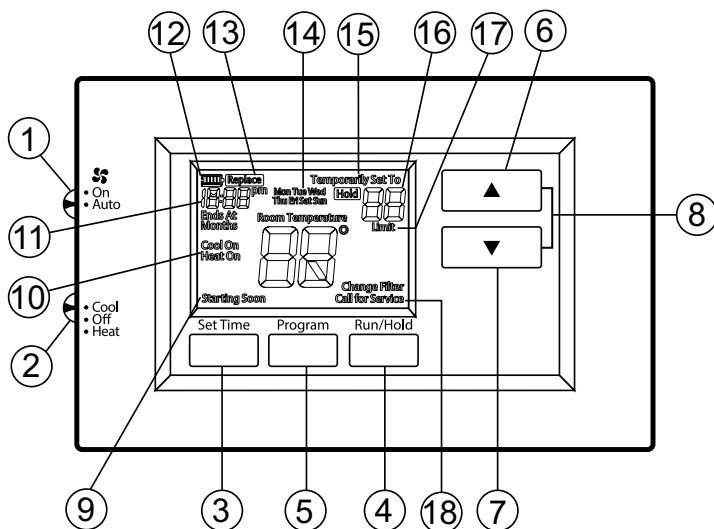
Do not allow the compressor to run unless the compressor oil heaters have been operational for 6 hours and the system has not been operational for at least 5 minutes.

USING THE THERMOSTAT

THERMOSTAT OVERVIEW

Before you begin using your thermostat, you should be familiar with its features, display and the location/operation of the thermostat buttons and switches.

THERMOSTAT BUTTONS AND SWITCHES	THE DISPLAY
1.) Fan Switch	9.) Thermostat is protecting the equipment from short cycling (5-minute delay)
2.) System Switch	10.) Indicates that the system is running in cool or heat
3.) Set Correct Time	11.) Displays the current time
4.) Hold a Permanent Temperature or Run Program	12.) Battery status indicator
5.) Access Programming	13.) Low battery indicator
6.) Raises Temperature Setting	14.) Day of the week used when programming a schedule
7.) Lowers Temperature Setting	15.) Permanent hold (bypassing the schedule)
8.) Access Menu Options	16.) Temperature setpoint
	17.) Displays when the thermostat setpoint has reached the maximum or minimum setting.
	18.) SEE TROUBLESHOOTING



Whenever "Replace" appears in the display, new premium brand AA alkaline batteries should be installed. If the house will be unoccupied for an extended period and either "Replace" or "Replace" is displayed, install new batteries before leaving.

THERMOSTAT OPERATION

Set Current Time and Day

Note: Time icons will flash at initial power up or after a reset.

- 1.) Press **Set Time**
- 2.) Use ▲ or ▼ to adjust the hour
- 3.) Press **Set Time** to advance to set the minutes and day of the week
- 4.) Press **Run/Hold** when finished.

The default program is **5-1-1 Day**, but can be setup as a **7-Day** or **Non-Programmable** thermostat (refer to the **Installer Menu** above)

- **Hold Temperature (bypassing the schedule)** – With the **System** switch set to **Heat** or **Cool**, momentarily press the **Hold** button. **Hold** will be displayed. Use ▲ or ▼ to adjust the temperature. The thermostat will hold the room temperature at the selected setting until you press **Run** to start program operation again.
- **Program Override (Temporary Hold)** – Press ▲ or ▼ until the desired temperature is displayed. The thermostat will override the schedule until the next programmed time period with a minimum override of 2 hours. Then the thermostat will automatically revert

THERMOSTAT SCHEDULE / PROGRAMMING

 **THERMOSTAT
HELPER**

For tips on programming your thermostat to receive optimal comfort, convenience and energy savings go to: thermostathelper.com

Energy Saving Factory Schedule

This thermostat is programmed with the energy saving settings shown in the table below for all days of the week.

	P1/Wake	P2/Leave	P3/Return	P4/Sleep
Heating Schedule	6:00 AM - 70°F	8:00 AM - 62°F	5:00 PM - 70°F	10:00 PM - 62°F
Cooling Schedule	6:00 AM - 78°F	8:00 AM - 85°F	5:00 PM - 78°F	10:00 PM - 82°F

Note: Thermostat can be programmed on or off the subbase

ENTERING YOUR PROGRAM

Set Current Time and Day

- 1. Press **TIME** button once. The display will show the hour only.

EXAMPLE: 

- 2. Press and hold either ▲ or ▼ until you reach the correct hour and AM/PM designation (AM begins at midnight; PM begins at noon).
- 3. Press **TIME** once again. The display window will show the minutes only.

EXAMPLE: 

- 4. Press and hold either ▲ or ▼ until you reach the correct minutes.
- 5. Press **TIME** once again. The display will show the day of the week.
- 6. Press ▲ or ▼ until you reach the current day of the week.
- 7. Press **RUN/HOLD** once. The display will show the correct time, day, room temperature and set-point temperature.

Enter Heating Program

- 1. Move the **SYSTEM** switch to **HEAT**.
- 2. Press **PRGM** once. "Mo Tu We Th Fr" (indicating weekday program) will appear in the

display. Also displayed are the currently programmed start time for the 1st heating period and the currently programmed temperature (flashing).

This display window shows that for the 1st weekday period, the start time is 6:00 AM, and 70° is the programmed temperature (this example reflects factory preprogramming).

3. Press ▲ or ▼ to change the displayed temperature to your selected temperature for the 1st heating program period.
4. Press PROGRAM once (the programmed time will flash). Press ▲ or ▼ until your selected time appears. The time will change in 15 minute increments. When your selected time is displayed, press PROGRAM again to return to the change temperature mode.
5. Press PROGRAM once. The currently programmed start time and setpoint temperature for the 2nd heating program period will appear.
6. Repeat steps 3 and 4 to select the start time and heating temperature for the 2nd heating program period.
7. Repeat steps 3 through 5 for the 3rd and 4th heating program periods.
8. Press PROGRAM once. "SA" (indicating Saturday program) will appear in the display, along with the start time for the 1st heating period and the currently programmed temperature.
9. Repeat steps 3 through 7 to complete Saturday heating programming.
10. Press PROGRAM once to change to SU (Sunday) heating programming and repeat steps 3 through 7 to complete Sunday programming.
11. When you have completed entering your heating program, press RUN/HOLD.

Enter Cooling Program

1. Move SYSTEM switch to COOL position.
2. Follow Enter Heating Program for entering your cooling program, using your selected cooling times and temperatures.

CAUTION

If the outside temperature is below 50°F, disconnect power to the cooling system before programming. Energizing the air conditioner compressor during cold weather may cause personal injury or property damage.

TROUBLESHOOTING

Symptom	Possible Cause	Corrective Action
No Heat/ No Cool/ No Fan (common problem)	1.) Blown fuse or tripped circuit breaker 2.) Furnace power switch to OFF 3.) Furnace blower compartment door panel loose or not properly installed 4.) Loose connection to thermostat or system	1.) Replace fuse or reset breaker 2.) Turn switch to ON 3.) Replace door panel in proper position to engage safety interlock or door switch 4.) Tighten Connections
No Heat	1.) System Switch not set to Heat 2.) Loose connection to thermostat or system 3.) Heating System requires service or thermostat requires replacement	Verify thermostat and system wires are securely attached. Diagnostic: Set System Switch to Heat and raise the setpoint above room temperature. Within five minutes the thermostat should make a soft click sound and "Heat On" should appear on display. This sound indicates the thermostat is operating properly. If the thermostat does not click, try the reset operation listed below. If the thermostat does not click after being reset, contact your heating and cooling service person or place of purchase for a replacement. If the thermostat clicks, contact the furnace manufacturer or a service person to verify the heating system is operating correctly.

(Troubleshooting continued on next page)

TROUBLESHOOTING (Continued)

Symptom	Possible Cause	Corrective Action
No Cool	1.) System Switch not set to Cool 2.) Loose connection to thermostat or system 3.) Cooling System requires service or thermostat requires replacement	Verify thermostat and system wires are securely attached. Diagnostic: Set System Switch to Cool and lower setpoint below room temperature. Same procedures as diagnostic for “No Heat” condition except set the thermostat to Cool and lower the setpoint below the room temperature. There may be up to a five minute delay before the thermostat clicks in Cooling if the compressor lock-out option is selected in the installer menu. (see INSTALLER MENU, item 50)
Heat, Cool or Fan Runs Constantly	Possible short in wiring, thermostat, heat, cool or fan system	Check each wire connection to verify they are not shorted or touching other wires. Try resetting the thermostat. If the condition persists contact your HVAC service person.
Thermostat Display & Thermometer Disagree	Thermostat display requires adjustment	Display can be adjusted +/-5°. See User Menu item 04
Furnace (Air Conditioner) Cycles Too Fast or Slow (narrow or wide temperature swing)	The location of the thermostat and/or the size of the Heating System may be influencing the cycle rate	Digital thermostats provide precise control and cycle faster than older mechanical models. The system turns on and off more frequently but runs for a shorter time. If you would like to increase cycle time, choose SLO for slow cycle in the Installer menu. (Reference menu items 30 & 35) If an acceptable cycle rate is not achieved, contact your HVAC service person.
“Call for Service” icon appears on display	1.) Heating system is not able to heat the space to within 10 degrees of the setpoint within 2 hours 2.) Cooling system is not able to cool the space to within 10 degrees of the setpoint within 2 hours 3.) If “--” is displayed for the Room Temperature, a replacement thermostat is needed 4.) None of the buttons operate on the thermostat	1.) See corrective action for “No Heat” 2.) See corrective action for “No Cool” 3.) Replace thermostat 4.) Make sure keypad lockout is not turned on (denoted by  icon)

Resetting the Thermostat or Thermostat Settings

If the thermostat has good batteries, but has a blank display or does not respond to key presses, the thermostat should be reset by removing the batteries for 2 minutes. This reset will not change the menu settings or program. If the condition persists after reinstalling the batteries, replace the thermostat.

To conveniently reset only the schedule and user settings back to factory defaults, press ▲, ▼ and **Set Time** buttons at the same time and hold until the display goes blank and resets.

WARNING: This product contains a chemical known to the state of California to cause cancer and birth defects and other reproductive harm.

HOMEOWNER HELP LINE: 1-800-284-2925